INSTALLATION GUIDE

Agent Monitor

for BES Clients

Installation Guide

Document Revision	July 16, 2026
Author	Mike Consuegra

Legal Notices and Trademarks

Third-Party Trademarks and Proprietary Rights

This document references products, product names, brand names, and trademarks owned by parties other than the author of this software. These references are provided for identification, interoperability, and educational purposes only. No claim of ownership, sponsorship, endorsement, or affiliation with these third parties is made or implied by their inclusion.

Product / Mark	Owner	Nature of Reference in This Document
HCL BigFix®, BigFix®, BigFix Platform™	HCLSoftware	Agent Monitor is designed to integrate with HCL BigFix via its published REST API and via standard BigFix Fixlet deployment. No source code from HCL BigFix is included in or distributed with this product.
IBM®	International Business Machines Corporation	Any references to IBM pertain to the historical evolution of the BigFix platform under prior ownership. No present-day affiliation with IBM is claimed.
PostgreSQL®	The PostgreSQL Global Development Group	PostgreSQL is an open-source relational database management system. No affiliation with or endorsement by the PostgreSQL Global Development Group is claimed.
Ubuntu®, Canonical®	Canonical Ltd.	Referenced as the operating system platform on which the Agent Monitor dashboard appliance is deployed. No affiliation with or endorsement by Canonical Ltd. is claimed.
Nginx™	F5, Inc.	Referenced as the reverse proxy component in the dashboard deployment architecture. No affiliation with or endorsement by F5, Inc. is claimed.
Python™	Python Software Foundation	Referenced as the programming language and runtime used to build the dashboard server and the Health Agent. No affiliation with or endorsement by the Python Software Foundation is claimed.
Windows®, Microsoft®	Microsoft Corporation	Referenced in the context of the Health Agent component, which runs as a Windows scheduled task on Windows endpoints. No affiliation with or endorsement by Microsoft Corporation is claimed.

General Disclaimer

All trademarks, registered trademarks, service marks, and product names mentioned in this document are the property of their respective owners. Where such marks appear without the ® or ™ symbol, the absence should not be interpreted as a statement that the mark is not protected.

Agent Monitor for BES Clients is an independent tool and is not a product of HCLSoftware, IBM, Microsoft, Canonical, F5, the PostgreSQL Global Development Group, the Python Software Foundation, or any other party referenced in this document.

® indicates a registered trademark. ™ indicates a trademark. All other marks are the property of their respective owners.

Table of Contents

What You Receive

You will receive the dashboard tarball plus the Health Agent bundles for the platforms you intend to monitor. The optional DMZ Forwarder is only needed when some endpoints cannot reach the dashboard directly.

File	Purpose	Install On
AgentMonitor-<version>.tar.gz	Dashboard appliance: web UI, ingest API, PostgreSQL schema, installer	Ubuntu Server (dedicated host)
agentmonitor-linux-<version>.tar.gz	Health Agent for Linux endpoints	Linux host with BES Client
agentmonitor-windows-<version>.zip	Health Agent for Windows endpoints	Windows host with BES Client
AgentMonitorForwarder-Setup.exe	DMZ Forwarder (optional relay)	Windows DMZ relay host

How agents enroll. The dashboard generates a small signed file called a charter (dashboard.charter) that you download once from the web UI and place next to the agent installer. The installer reads it, verifies it, and enrolls the endpoint with nothing to type by hand. The charter is created in Part 2 and used in Part 3.

Architecture Overview

The dashboard holds the endpoint history, the tenant configuration, and the web UI. The Health Agent runs on each endpoint, performs a small set of local checks, and reports the results back to the dashboard. The optional DMZ Forwarder relays agent traffic inward for endpoints that sit in networks that cannot reach the dashboard directly. The diagram below illustrates how the components interact across your environment.

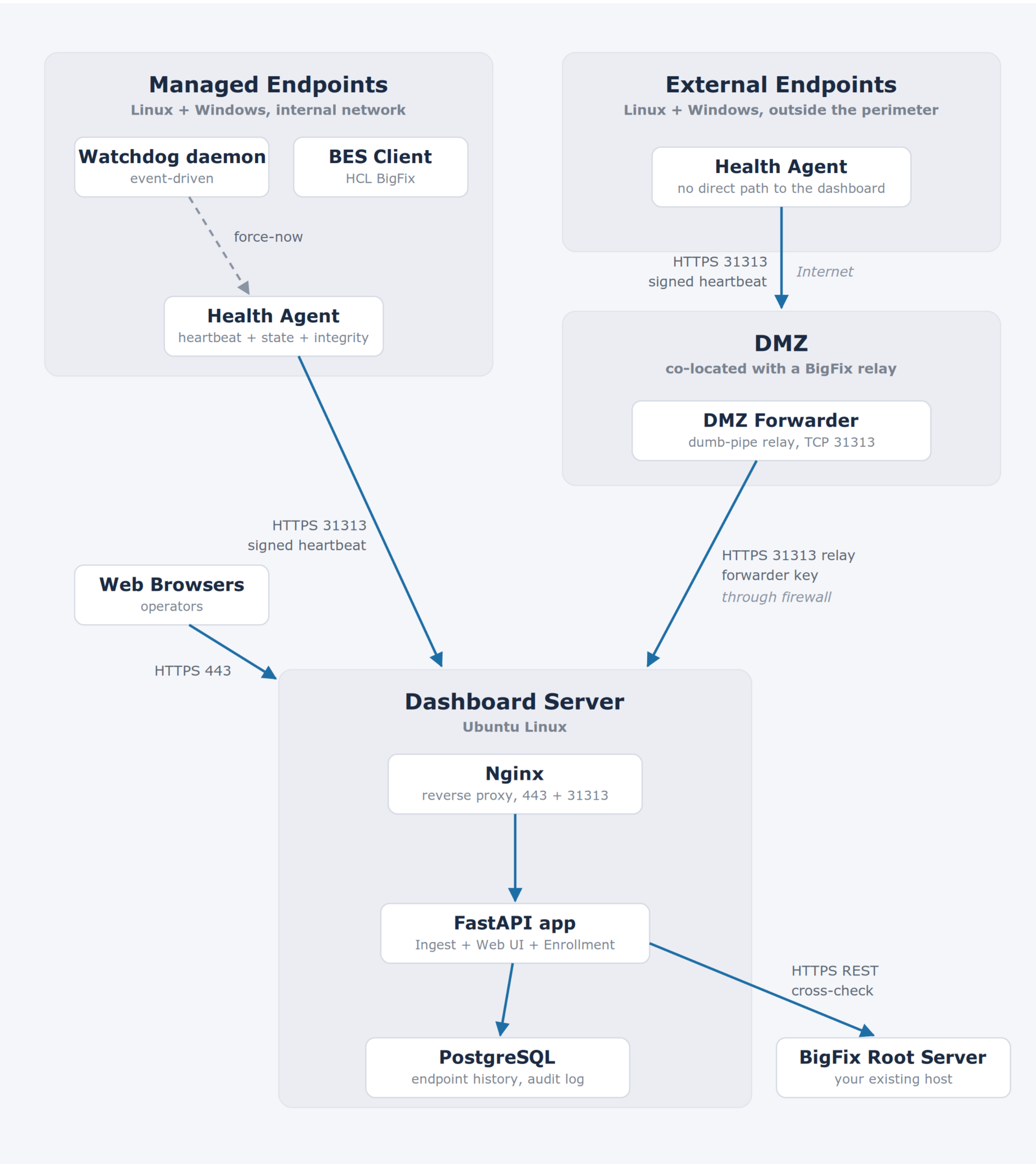


Figure 1: Agent Monitor system architecture

Data Flow

- **Health Agent** The agent runs on a schedule on each endpoint (a systemd timer on Linux, a scheduled task on Windows) and posts a heartbeat to the dashboard. Every heartbeat is signed with a per-endpoint keypair; the default algorithm is post-quantum ML-DSA-65.
- **Watchdog** A watchdog process on each endpoint reacts to the BES Client stopping or restarting and triggers an immediate agent run. On Linux it is event-driven and reacts within seconds; on Windows it polls the Service Control Manager on a dashboard-tunable interval (60 seconds by default).
- **BigFix REST cross-check** On a schedule the dashboard calls the BigFix REST API and reconciles what BigFix sees against what the Health Agent reports. This is how it distinguishes a genuinely silent agent from an endpoint BigFix no longer sees, and how it populates the Disagreement card. An endpoint that stops reporting is marked silent after 24 hours by default, and promoted to Disagreement after 72 hours if BigFix still corroborates it. Both thresholds are configurable.
- **Web UI** Operators use any modern browser over HTTPS on port 443.
- **Report encryption (Strongbox)** Reports can optionally be encrypted end to end so a middle man cannot read them. This is a per-tenant policy that ships off by default.
- **Boot beacon** On the first agent run after a reboot, the agent sends a one-time boot beacon so a mass reboot clears any false silent state quickly without overwhelming the server.
- **Self-check cadence** Between watchdog-triggered runs the agent self-checks on an internal interval that defaults to 15 minutes and is tunable per endpoint from the dashboard. Once every 24 hours it sends a liveness heartbeat even if nothing has changed, so a quiet-but-healthy endpoint never drifts into the silent bucket.
- **DMZ Forwarder** The optional forwarder is a dumb pipe. It accepts agent reports and relays them inward to the dashboard, never reading or altering report contents. Agents given a forwarder address try the dashboard directly first, fall back to the forwarder, and spool locally if neither is reachable.

Ports

The key ports used by the system:

- Operator web UI (browser to dashboard): 443 (HTTPS).
- Agent and forwarder traffic to the dashboard: TCP/31313 (HTTPS). This is the platform's dedicated machine-to-machine port and the only port the dashboard serves agent traffic on; the port is fixed, not configurable. The operator web UI is separate, on 443.
- Agent traffic to the DMZ Forwarder (inbound): TCP/31313.
- Dashboard to BigFix REST API: 52311 (default).
- Dashboard application (internal): 127.0.0.1:5001, loopback only, the Nginx upstream.

Prerequisites

Dashboard Server (Ubuntu)

Requirement	Details
Operating System	Ubuntu Server 24.04 LTS or 26.04 LTS (amd64)
Access	Root or sudo
RAM	Minimum 8 GB. 16 GB for 25,000–100,000 agents. 32 GB and a dedicated PostgreSQL data partition above 100,000, or for heavy concurrent operator queries or very short self-check intervals.
Disk	Minimum 20 GB free on the partition that holds /opt and /var
Disk (Large Environment)	Add ~1MB of data-partition headroom per additional endpoint beyond 25,000. Environments above 50,000 should plan on 100GB or a dedicated PostGreSQL data partition.
Network (outbound)	Internet access during installation (for apt packages)
Network (REST)	Must reach the BigFix Root Server REST API
Network (ingest)	Must be reachable from every endpoint on TCP/31313
Ports	443 (HTTPS) to users; TCP/31313 (HTTPS) from endpoints and forwarders.

Hands-off setup. The installer brings up PostgreSQL, Nginx, Python, and all dependencies. You don’t need to pre-install anything beyond base Ubuntu Server image although it does detect if you already installed them.

Managed Endpoints

Platform	Versions
Linux	Ubuntu 22.04 / 24.04 / 26.04, RHEL 8 / 9 (systemd required)
Windows	Windows 10 / 11, Windows Server 2016 / 2019 / 2022 / 2025
macOS	Not included in this release (planned for a future release)

The endpoint must already be enrolled in your BigFix environment (the BES Client must be installed and reporting). Agent Monitor does not replace the BES Client; it watches it.

BigFix Operator Account

The dashboard requires a BigFix operator account to perform its cross-check. This account **must be a Master Operator** (MO). A non-MO account will produce incomplete cross-check results.

> **Why Master Operator.** The cross-check relies on the BigFix REST API returning the full Computer list plus **Last Report Time** and online status for every endpoint, regardless of which BigFix site it belongs to. Only the Master Operator role has that scope.

Part 1: Dashboard Server Installation (Ubuntu)

These steps install the dashboard appliance on a clean Ubuntu Server host. The installer is interactive and prompts for the values it cannot guess.

Step 1: Transfer the Dashboard Package

Transfer the dashboard tarball to your Ubuntu server. SCP is the simplest path:

From your workstation

```
scp AgentMonitor-<version>.tar.gz user@your-server:/tmp/
```

Step 2: Extract the Package

On the server

```
ssh user@your-server
cd /tmp
tar -xzf AgentMonitor-<version>.tar.gz
cd AgentMonitor-<version>
```

You should see the following layout:

Tarball layout

```
AgentMonitor-<version>/
  app/                           # Application code (FastAPI routers, models,
services)
  static/                        # Web frontend assets (JS, CSS, images)
  scripts/                       # Compile and helper scripts
  migrations/                    # Versioned schema migrations
  install.sh                     # Automated installer
  database.sql                   # Initial schema
  nginx-bigfix-health-archive    # Nginx site template
  requirements.txt               # Python dependencies
```

Step 3: Run the Installer

bash

```
sudo bash install.sh
```

The installer prints a banner and begins the installation in stages.

```
==============================================================================
  Agent Monitor for BES Clients — Installer
  Version 2.6.5
==============================================================================


[20:50:55] === Pre-flight Checks ===
  Detected: Ubuntu 26.04 (resolute)

[20:50:55] === Prerequisite Check ===
  Found: curl
  Found: python3
  Found: openssl
  Found: free
  Found: lsblk
  Found: nproc
  Found: useradd
  Found: find
  All prerequisites satisfied.
  Python 3.14 OK (>=3.12).
  Memory: 7424MB available
  Disk:   14689MB free on /opt

[20:50:56] === Configuration ===

  Provide the following. Press Enter to accept defaults in [brackets].

  --- Web Admin Account ---
  Admin username [admin]:
```

Step 4: Enter Admin Account Details

The installer prompts for the initial administrator account:

- **Admin username** The username you will use to log into the web UI (for example, `admin`)
- **Admin password** Entered twice for confirmation. Choose a strong password; it is stored hashed.
- **Admin email** Optional. Required for password reset and admin email notifications.

```
[20:50:56] === Configuration ===

  Provide the following. Press Enter to accept defaults in [brackets].

  --- Web Admin Account ---
  Admin username [admin]:
  Admin password: ********
  Confirm Admin password: ********
  Admin email (optional, press Enter to skip):
  WARNING: No admin email set. Password reset and email notifications for the admin ac
count will be unavailable until you set one via the web UI (Administration > User Mana
gement).
```

Step 5: Database User

The installer creates a dedicated PostgreSQL role for the application and generates its password automatically (the password is encrypted into the configuration file). You are only asked for the role name; press Enter to accept the default (bfhealthapp) unless you have a naming convention to follow.

Step 6: SSL Certificate Configuration

By default the installer generates a self-signed certificate. It then asks whether you want to use your own certificate instead:

- **Answer N (the default)** to use the generated self-signed certificate. Browsers show a one-time warning you can dismiss. Recommended for first-time installs and internal use.
- **Answer y** to supply the paths to your own certificate (.crt) and private key (.key). The installer copies them into place and configures Nginx to use them. Use this in production.

```
--- Database ---
Database password: [auto-generated, will be encrypted in .env]
Database user [bfhealthapp]:

--- SSL Certificate ---
A self-signed certificate will be generated by default.
Do you have a CA certificate that you would like to use instead? (y/N) [N]:
```

Step 7: Agent Reporting Address

The installer asks for the URL that endpoints will use to reach this dashboard. This value becomes the default ingest URL baked into the charter file. The port is fixed at the platform port 31313 and cannot be changed: the suggested default is https://<this host's address>, and the installer normalizes whatever you enter onto :31313. Set the host part to how endpoints actually reach the dashboard (a stable DNS name is preferred; an IP works if it never changes).

```
--- Agent Reporting Address ---
The address agents report to, baked into every charter you generate.
WARNING: This value must be reachable by EVERY endpoint across your network.
WARNING: Use a stable DNS name where possible; an IP works only if it never changes.
Dashboard URL for agents [https://192.168.0.46:31313]:
```

Step 8: Dependency Check

Since there are certain dependencies that have to be downloaded, the installer will check whether the versions about to be downloaded are the same as the ones the platform was built with. If it detects that the dependency versions are newer it will give you the option to install the exact versions used in the build or the newest versions available.

```
============================================================
  Dependency version check
============================================================
This release was built and tested with specific Python library
versions. Your system would now install newer ones for:

  anyio                  tested 4.14.0         ->   available 4.14.2
  APScheduler            tested 3.11.2         ->   available 3.11.3
  cffi                   tested 2.0.0          ->   available 2.1.0
  charset-normalizer     tested 3.4.7          ->   available 3.4.9
  click                  tested 8.4.1          ->   available 8.4.2
  fastapi                tested 0.138.0        ->   available 0.139.2
  greenlet               tested 3.5.2          ->   available 3.5.3
  pillow                 tested 12.2.0         ->   available 12.3.0
  typing_extensions      tested 4.15.0         ->   available 4.16.0
  tzlocal                tested 5.4.3          ->   available 5.4.4
  uvicorn                tested 0.49.0         ->   available 0.51.0

These newer versions were not tested with this build. Choose:

  [T] Install the exact tested versions   (recommended for production)
  [C] Continue with the latest versions    (not tested with this build)
  [A] Abort the installation

Your choice [T/C/A]:
```

Step 9: Performance Tuning

The installer will auto detect the host's hardware specs and make a recommendation about tuning the database and kernel :

- **PostgreSQL:** The installer will make recommendations about tuning PostgreSQL for performance
- **Kernel Virtual Memory:** The installer will recommend a memory management setting

```
[03:11:06] === PostgreSQL Performance Tuning ===

  Detected hardware:
    Total RAM:    7424 MB (~7 GB)
    CPU cores:    4
    PG storage:   SSD

  Recommended PostgreSQL settings for this server:
    shared_buffers                   = 1856MB
    effective_cache_size             = 4454MB
    work_mem                         = 16MB
    maintenance_work_mem             = 256MB
    random_page_cost                 = 1.1
    effective_io_concurrency         = 200
    max_worker_processes             = 4
    max_parallel_workers             = 4
    max_parallel_workers_per_gather  = 2
    max_parallel_maintenance_workers = 2
    wal_buffers                      = 16MB
    checkpoint_completion_target     = 0.9

  Plus kernel: vm.swappiness = 10

  Apply these PostgreSQL tuning settings now? [Y/n]:
```

It is recommended to accept the tuning parameters for better application performance.

The installer then runs through its stages automatically:

1. Install system packages (PostgreSQL, Nginx, Python, build tools)
2. Create the application user (bfhealthapp) and the directory tree
3. Set up the Python virtual environment and install dependencies
4. Deploy the application files and install the integrity manifest
5. Initialize the database, load the schema, and apply all migrations
6. Configure SSL and Nginx (HTTP redirects to HTTPS on 443) and write the encrypted configuration file
7. Create and start the two systemd services: the API service (bigfix-health-archive) and the scheduler service (bigfix-health-archive-scheduler)
8. Start the application and run a health check

When complete, you will see a success message with the dashboard URL and the service status reading "active (running)."

```
============================================================================
  Installation Complete! (v2.6.5)
============================================================================

  Access the application:
    URL:      https://192.168.0.46
    Username: admin
    Password: (as configured during install)

  Service management:
    Status:   sudo systemctl status bigfix-health-archive bigfix-health-archive-scheduler
    Restart:  sudo systemctl restart bigfix-health-archive bigfix-health-archive-scheduler
    API log:  sudo journalctl -u bigfix-health-archive -f
    Sched log: sudo journalctl -u bigfix-health-archive-scheduler -f

  Important paths:
    App:      /opt/bigfix-health-archive/
    Config:   /opt/bigfix-health-archive/config/.env
    Data:     /opt/bigfix-health-archive/data/diagnostics/
    Backups:  /opt/bigfix-health-archive/backups/


  This is a fresh install. Next steps:
    • If migrating from another Agent Monitor server:
        1. Log in here with the throwaway admin credentials you just set
        2. Administration → System Settings → Backup & Restore
        3. SERVER MIGRATION → IMPORT MIGRATION, upload your .bch-mig
        4. After import completes, restart both services and log in with
           the SOURCE server's admin credentials (yours will be overwritten)
    • If standing up a new deployment:
        Administration → System Settings → Connections to configure BigFix REST
        Administration → API Keys to mint agent credentials

  Self-signed certificate — your browser will show a warning on first
  visit. Replace via System Settings → Certificates when ready.
```

Warmup window. After a restart or a fresh install, the dashboard takes roughly 30 to 60 seconds to warm up under load. If your first browser hit returns a 502 Bad Gateway, wait 60 seconds and refresh. A real failure shows up in the application log and in systemctl status.

Part 2: First Login and Configuration

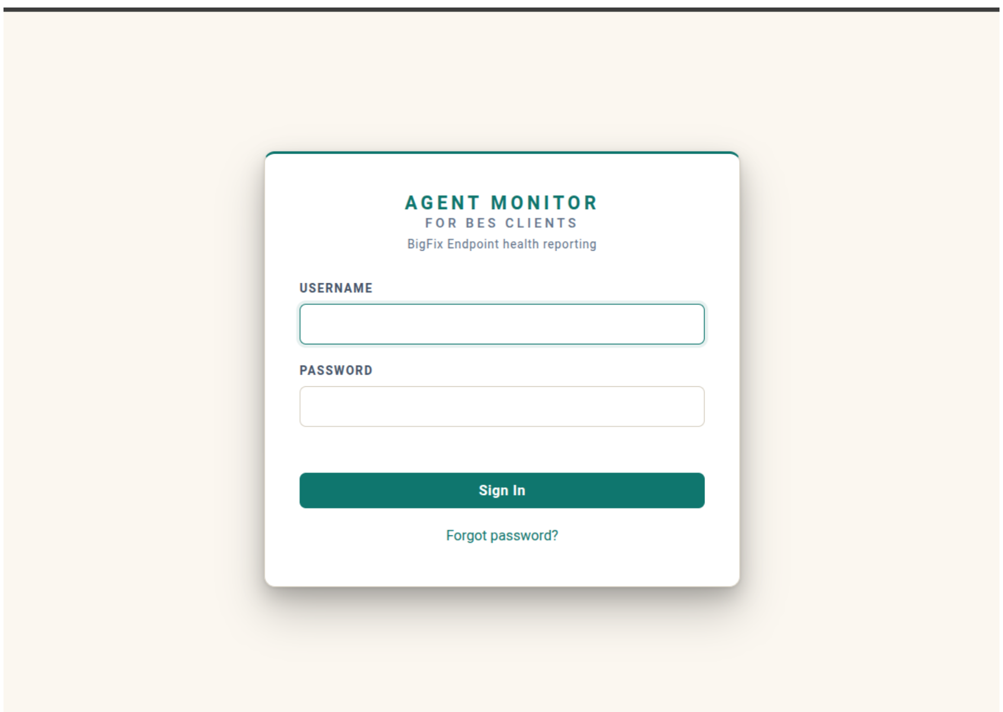
After the installer finishes, log into the web UI and complete the one-time configuration: verify the BigFix connection, set your report encryption policy if you want it, and generate the charter you will use to deploy agents. You do not create tenants by hand; a tenant is created automatically the first time an endpoint reports in, keyed to the BigFix environment it belongs to.

Step 1: Open the Web UI

In your browser

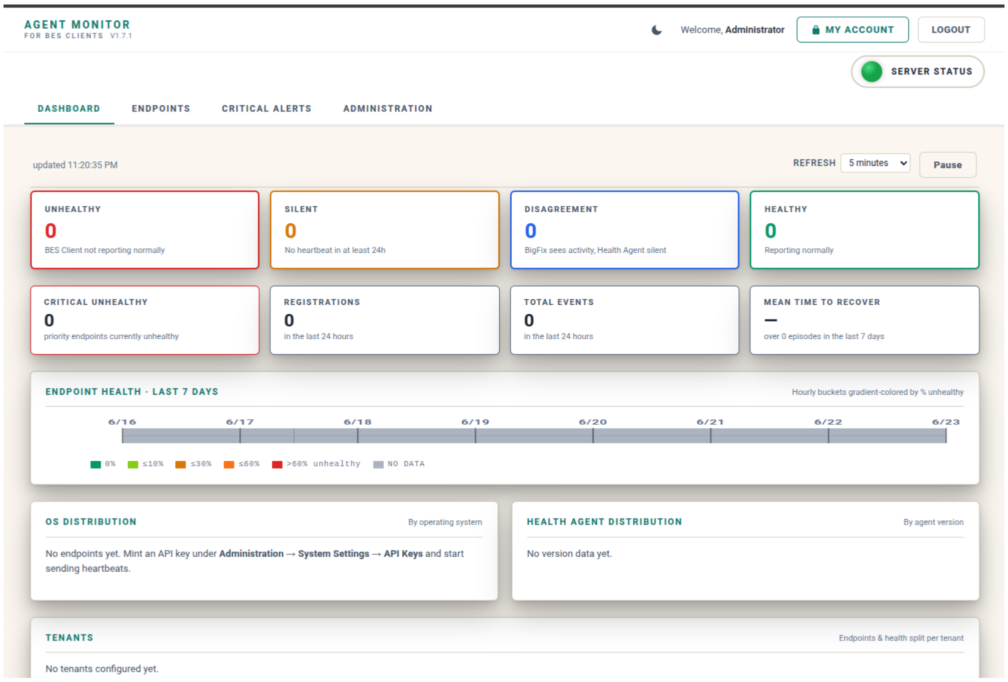
```
https://your-server-ip
```

If you used the self-signed certificate option in Part 1, your browser will show a security warning the first time. Click through to proceed. The connection is still encrypted, and the warning will disappear once you replace the certificate with one your browser trusts.



Step 2: Log In

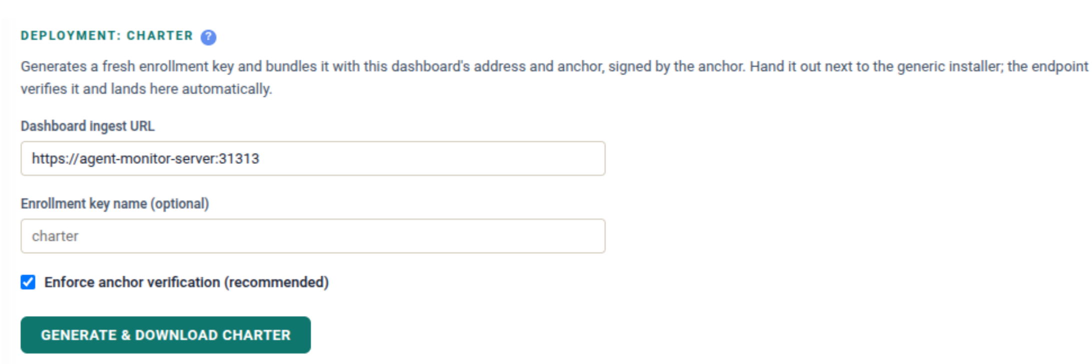
Enter the admin username and password you created during installation. The first login lands you on the Dashboard tab, which will show zeros for every counter because no endpoints have reported yet.



Step 3: Generate and Download the Charter

The charter is the single file you hand to the agent installer. It carries the dashboard ingest URL, a new enrollment key, the anchor fingerprint the agent pins its trust to, your encryption enforcement setting, and (when a DMZ Forwarder exists) the forwarder trust bundle.

To generate it, go to Administration > System Settings > Agent Builds. In the DEPLOYMENT: CHARTER section, confirm the Dashboard ingest URL, optionally set an Enrollment key name, leave Enforce anchor verification (recommended) checked, and click GENERATE & DOWNLOAD CHARTER. Your browser downloads a file named dashboard.charter.

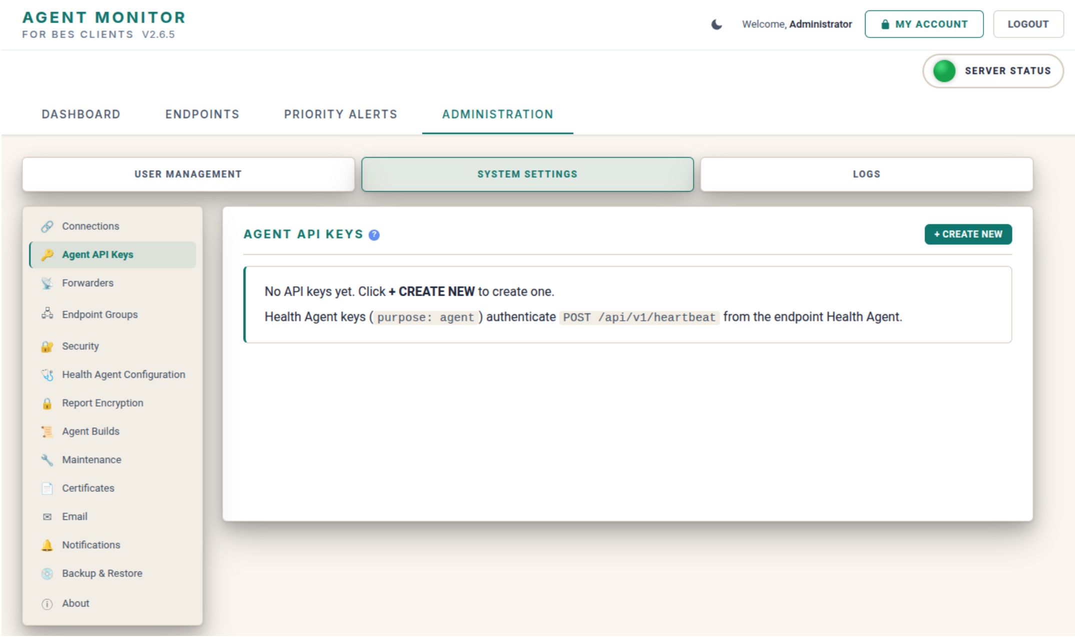


> **Keep the charter file safe.** The enrollment key exists only inside the charter and is never stored on the dashboard in the clear. One charter serves your whole environment; the tenant an endpoint belongs to is determined automatically when it first reports.

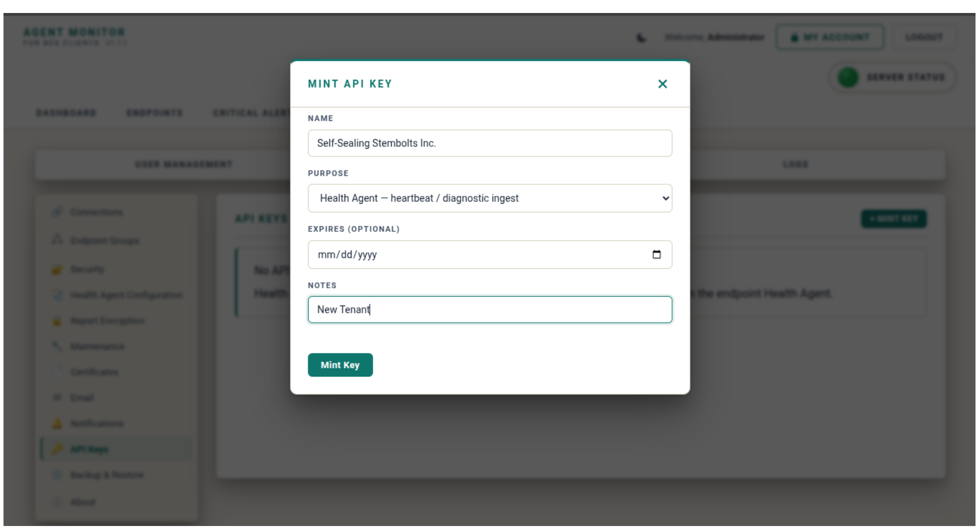
Step 4: Create an Agent API Key (Optional)

Generating a charter in Step 6 creates its own enrollment key automatically, so a separate API key is only needed if you want a named key for integrations or your own tracking. To create one:

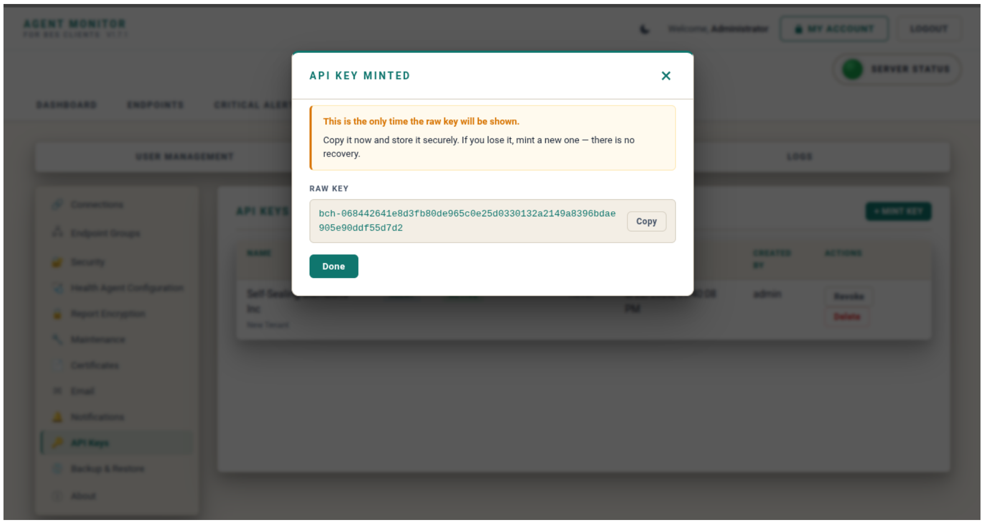
1. Administration > System Settings > Agent API Keys



2. Click + CREATE NEW and fill in Name, Purpose (Health Agent, Integration, or Admin), and the optional expiry and notes.



3. Click Create Key and copy the raw key (prefixed bch-). It is shown only once.

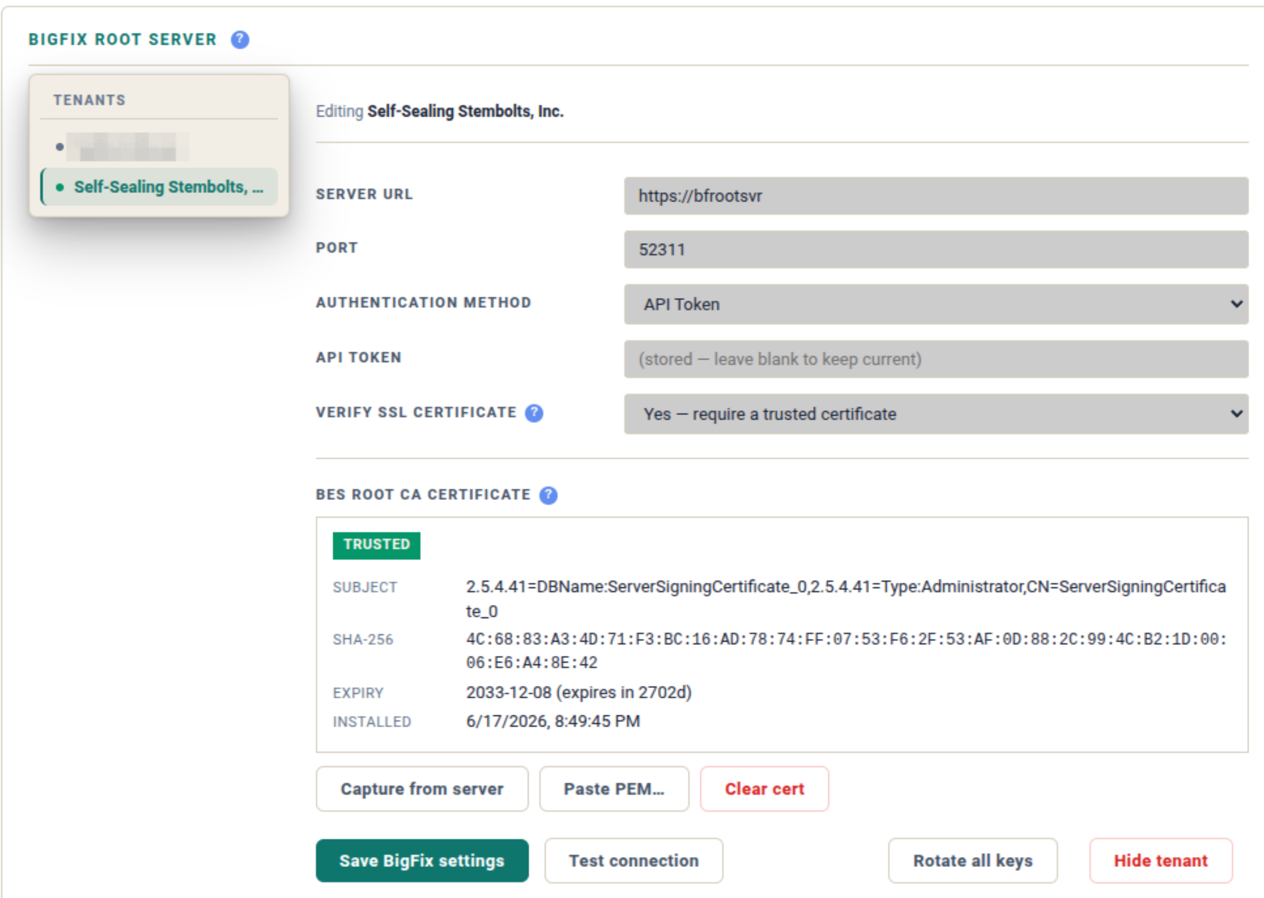


Note: The key will not be displayed again. Copy it somewhere that you will be able to retrieve it later when it's time to deploy agents via the BigFix Console.

Step 5: Configure the BigFix Connection and Root CA

Agent Monitor cross-checks each endpoint against the BigFix REST API. For that to work over HTTPS, the dashboard must trust your BigFix Root Server's certificate. Each tenant carries its own Root CA, so you can host several BigFix environments side by side.

1. Go to Administration > System Settings > Connections and select the tenant in the sidebar.
2. On the BIGFIX ROOT SERVER card, enter the server URL, port (default 52311), and Master Operator credentials. Under BES ROOT CA CERTIFICATE, choose Capture from server, or Paste PEM and paste your Root CA, then Install pasted cert.
3. The setting takes effect immediately.
4. The dashboard reports success or the exact reason it failed.

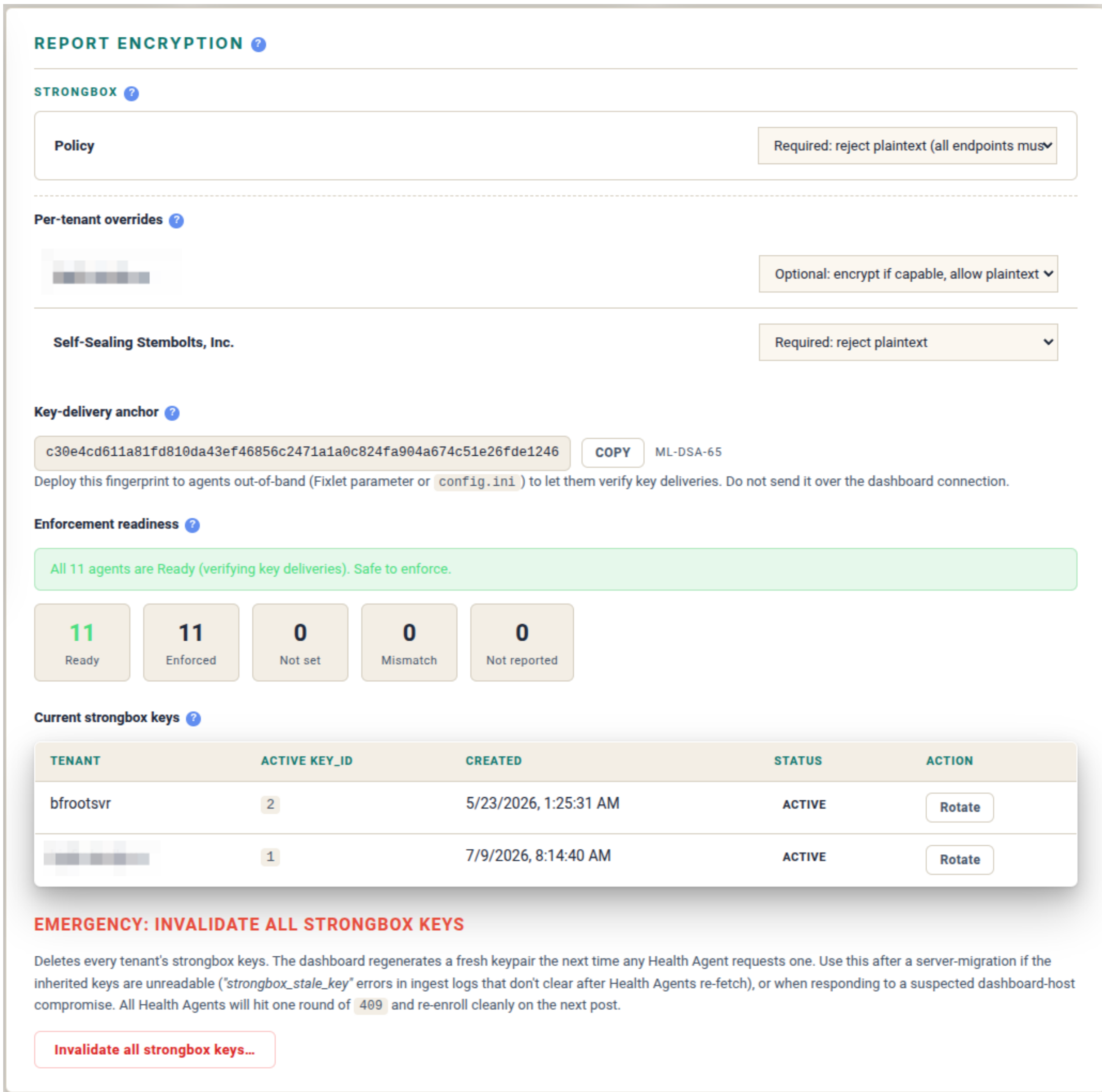


Validity-only check. The dashboard's cross-check is permissive on purpose. It will accept self-signed BigFix Root CAs and skip CN validation. The one thing it does reject is an expired certificate, since that fails closed in the trust chain.

Step 6: Set the Report Encryption Policy (Optional)

If you want diagnostic bundles encrypted end to end, set the policy now; otherwise skip this step and revisit it later. Go to Administration > System Settings > Report Encryption.

1. Set the Policy: Off (no encryption, the default), Optional (encrypt when the endpoint supports it, allow plaintext during a rolling migration), or Required (reject plaintext).
2. To vary it by tenant, use Per-tenant overrides, where each tenant inherits the global policy or sets its own.



Step 7: Stage the Agent Bundles on Your BigFix Console Workstation

The Health Agent installer reads the platform bundle and the charter. If you will deploy through BigFix, stage the bundle and the charter where the Console can read them:

Suggested layout on the operator workstation

```
C:\BigFix\agentmonitor\
agentmonitor-linux-<version>.tar.gz
agentmonitor-windows-<version>.zip
dashboard.charter
```

Part 3: Health Agent Installation

The Health Agent installs from a self-contained installer that reads the dashboard.charter you generated in Part 2. You can run it directly on an endpoint for a pilot, or deploy it at scale through a standard BigFix Fixlet. Both paths use the same installer and the same charter.

How the charter is used. Place dashboard.charter in the same folder as the installer, or pass it with the --charter option. The installer verifies the charter's signature before trusting anything in it. If the charter is missing or altered, verification fails closed and the installer falls back to asking for a dashboard URL and key.

Linux Health Agent

Copy the Linux bundle and the charter to the endpoint (or to a staging location a Fixlet will read), and extract the bundle:

```
tar -xzf agentmonitor-linux-<version>.tar.gz
cd agentmonitor
```

Run the installer as root, with the charter alongside it:

```
sudo bash install.sh
```

The installer auto-detects dashboard.charter file, verifies it, and writes the enrollment settings into the agent configuration. If no charter is present and the console is interactive, it prompts instead and live-validates the values against the dashboard before continuing:

```
1. Dashboard URL (e.g. https://192.168.0.45):
2. Agent API key (bch-...):
```

What the installer creates:

- Install directory: /opt/agentmonitor
- Configuration: /etc/agentmonitor/config.ini
- State and logs: /var/lib/bigfix-health-agent

- Service user: agentmonitor (system account, no shell)
- Units: agentmonitor.timer (backstop about every 10 minutes), agentmonitor.service, and agentmonitor-watchdog.service (event-driven)

Any older bigfix-health-agent units are stopped and removed automatically, and the endpoint's identity and history are preserved.

Windows Health Agent

Copy the Windows bundle to the endpoint and extract it. The self-contained installer, AgentMonitor-Installer.exe, is inside. Place dashboard.charter next to it, then run the installer as Administrator.

The installer bundles the signed agent executable and its signing root certificate, so it establishes its own code-signing trust; no separate trust-bootstrap step is required. It auto-detects and verifies the charter, then installs. Run interactively with no charter, it prompts for the dashboard URL and key and live-validates them, exactly as the Linux installer does. Run silently or through a Fixlet, it never prompts.

What the installer creates:

- Install directory: C:\Program Files\AgentMonitor
- Agent executable: AgentMonitor.exe
- Agent task: \AgentMonitor (runs about every 15 minutes)
- Watchdog task: \AgentMonitor-Watchdog (repeats about every 10 minutes and at startup)

Any older \BigFix-HealthAgent tasks are removed automatically.

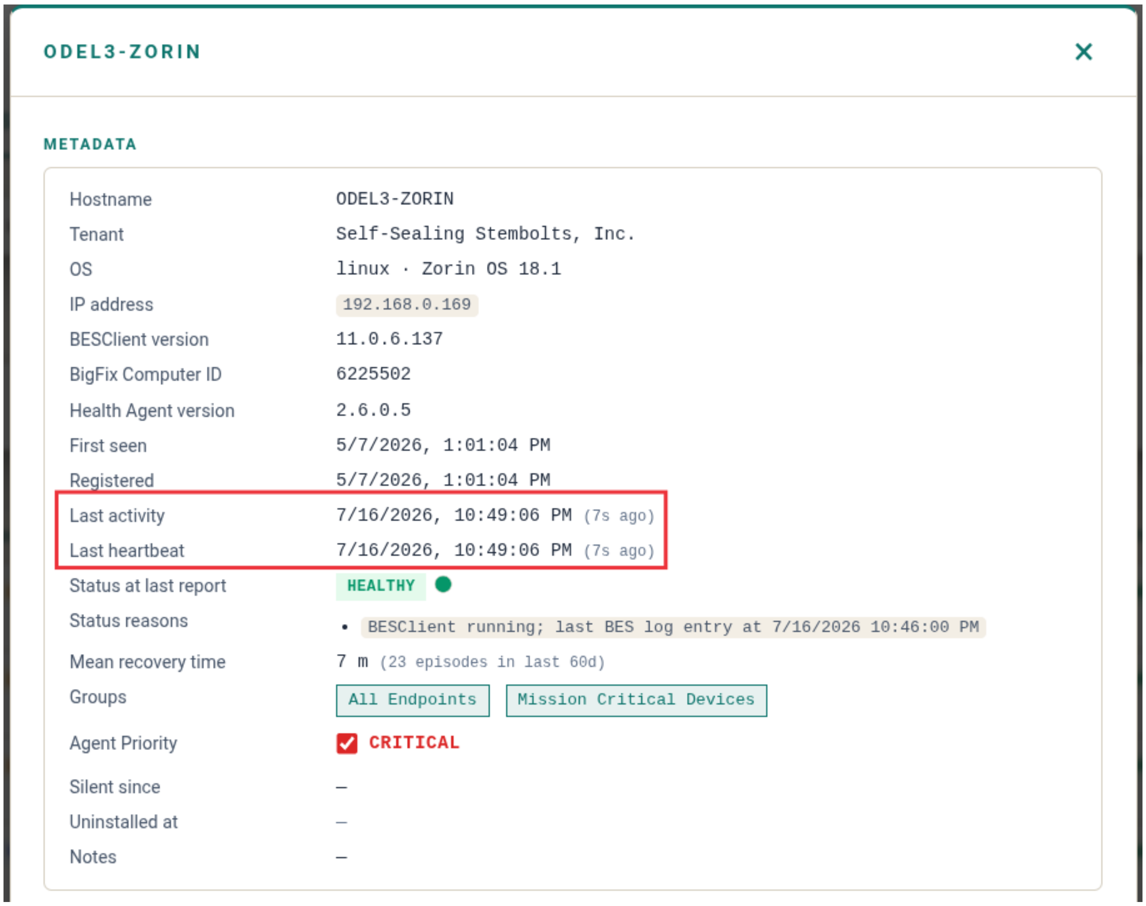
Deploying at Scale with BigFix

For a full rollout, deploy the same installer and charter through BigFix rather than touching endpoints by hand:

- Stage the platform bundle and dashboard.charter on the BES Root Server.
- Use a Fixlet to download both to the endpoint, extract, and run the installer non-interactively; it reads the charter and enrolls with no prompts.
- Target a small pilot computer group first (10 to 50 endpoints), confirm heartbeats and integrity badges appear, then expand.
- Because the installer is idempotent and version-aware, re-running it against current endpoints is safe.

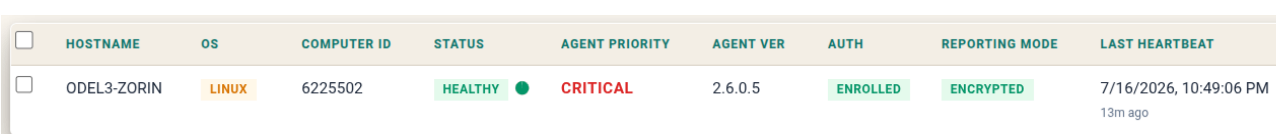
Confirm Endpoints Are Reporting

Return to the web UI, open the Endpoints tab, and open any newly enrolled endpoint. Its last-heartbeat timestamp should be within the past few minutes.



Confirm Integrity State

In the Endpoints list, integrity appears as a badge beside the health status, and only when something is wrong. A correctly installed agent shows no integrity badge at all: the clean state is silent. Three badges can appear: TAMPERED means the reported binary SHA-256 does not match the build attestation on file for that version and platform; CHAIN INVALID (Windows only) means the Authenticode signature does not chain to the Agent Monitor root certificate on the endpoint; UNKNOWN BUILD means no attestation is on file yet for that build. To confirm integrity after deployment, verify that newly enrolled endpoints show no integrity badge.

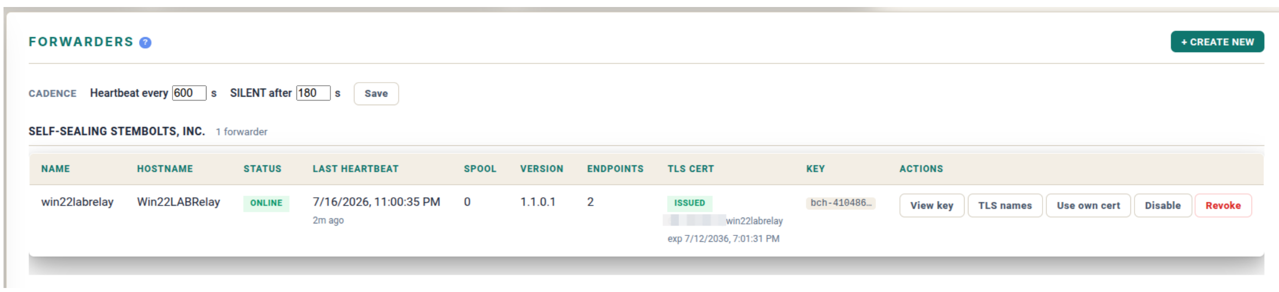


Part 4: DMZ Forwarder Installation (Optional)

Install a DMZ Forwarder only when some endpoints cannot reach the dashboard directly, for example endpoints served by an internet-facing BigFix relay. The forwarder sits in the exposed network, accepts agent reports on TCP/31313, and relays them inward to the dashboard. It is a dumb pipe: it never reads or changes report contents, and it authenticates itself to the dashboard with its own key.

Step 1: Register the Forwarder in the Dashboard

In the web UI, go to Administration > System Settings > Forwarders and register a new forwarder with a name. The dashboard creates a forwarder key (prefixed bch-) for it. Enter the TLS names that agents will use to dial this forwarder; these become the subject-alternative-names in the forwarder's signed certificate. You can enter DNS names or, when you have no resolvable name, an IP address (IPv4 or IPv6).



Bring your own certificate (optional). If you prefer your own PKI for the forwarder's certificate, switch the forwarder to your own CA and supply the certificate. Otherwise the dashboard issues and renews the forwarder's certificate automatically.

Step 2: Install the Forwarder on the DMZ Relay

On the Windows DMZ host, run AgentMonitorForwarder-Setup.exe. The self-contained installer stages the forwarder, validates the forwarder key against the dashboard, registers an auto-start Windows service, opens the Windows Firewall for inbound TCP/31313, starts the service, and confirms the listener is answering.

The values it needs are the upstream dashboard URL and the forwarder key from Step 1. Its configuration uses these sections and keys:

- [dashboard] api_url: upstream dashboard URL (for example https://dashboard.example.com:31313)
- [dashboard] verify_ssl: verify the dashboard's TLS certificate
- [forwarder] forwarder_key: the bch- forwarder key from Step 1
- [listener] bind, port: inbound bind address and port (default 0.0.0.0:31313)
- [tls] cert_mode: auto (dashboard-issued certificate) or byo (your own)

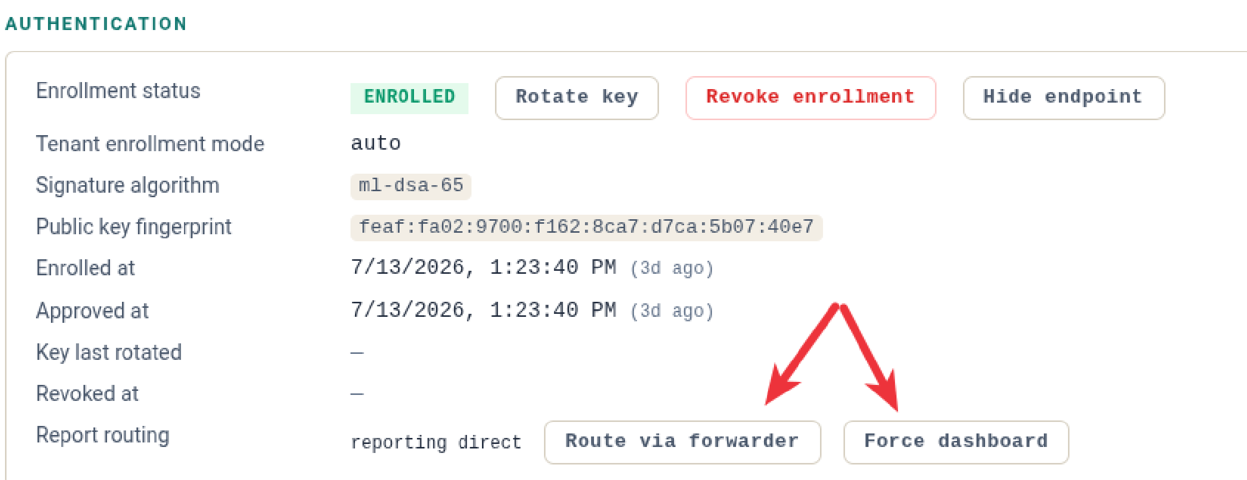
Step 3: Certificate Issuance (Automatic)

With cert_mode = auto, the forwarder generates a certificate signing request and submits it to the dashboard, which acts as a private certificate authority and returns a signed certificate carrying the TLS names you set in Step 1. Renewal is automatic and hot-swaps the running listener with no restart. Agents learn to trust this authority through the charter's forwarder trust bundle and a self-healing push from the dashboard, so no manual certificate distribution to endpoints is needed.

> **If the certificate is not yet issued.** If the dashboard cannot be reached or no TLS names are set yet, the forwarder self-signs so its listener still comes up, and that leg stays unverified until a proper certificate is issued. Set the TLS names in Step 1 and the forwarder picks up its signed certificate on its next check-in.

Step 4: Point Endpoints at the Forwarder

In most deployments no endpoint configuration is needed: agents automatically discover forwarder candidates from the BES Client's own relay settings and try them on the platform port when the dashboard is not reachable directly. You can steer routing in two ways. Per endpoint from the dashboard: open the endpoint's detail panel and use the routing control (Route via forwarder or Force dashboard)



The choice is applied on the endpoint's next heartbeat, with no Fixlet or config edit. At install time in the agent configuration: set [dashboard] forwarder_url to pin a specific forwarder, and optionally prefer_forwarder = true to route through it without probing the dashboard first. A dashboard routing choice overrides the config file default. Agents fall back to a direct dashboard connection when possible and spool locally if neither path is available.

Verification

After completing the parts you deployed, run through this checklist to confirm everything is working as expected.

Dashboard Server

1. **Services running.** Run sudo systemctl status bigfix-health-archive and sudo systemctl status bigfix-health-archive-scheduler; both should report active (running).
2. **Health endpoint.** Run `curl -sk https://localhost/api/health | python3 -m json.tool` and confirm it returns status: healthy with every sub-check passing.
3. **Web UI accessible.** Open `https://your-server-ip` in a browser and log in successfully.
4. **BigFix connection.** On the Connections page, confirm Test connection succeeds and the last cross-check is recent.

Health Agent (Linux endpoint)

1. **systemd timer exists.** Run `systemctl status agentmonitor.timer`
2. **Watchdog daemon running.** Run `systemctl status agentmonitor-watchdog.service`
3. **Identity preserved across upgrade.** If this endpoint was previously on the old bigfix-health-agent layout, confirm `/var/lib/bigfix-health-agent/identity.json` exists and the dashboard shows the same agent_id it had before. Endpoint history is preserved across the rename.

Health Agent (Windows endpoint)

1. **Scheduled tasks exist.** Open Task Scheduler and confirm `\AgentMonitor` and `\AgentMonitor-Watchdog` are present and enabled.
2. **Installed binary visible.** Confirm `C:\Program Files\AgentMonitor\AgentMonitor.exe` exists. Right-click and choose Properties to confirm File Version reads <version> and the Digital Signatures tab shows a valid signature.
3. **Integrity state clean. In the web UI, the endpoint shows no integrity badge beside its health status; a badge appears only when something is wrong.**

DMZ Forwarder (if installed)

- Service running: the Agent Monitor Forwarder Windows service is running.
- Listener up: the forwarder answers on TCP/31313.

- Seen by the dashboard: the forwarder appears on the Forwarders page with a recent heartbeat, and its certificate shows as issued.

End-to-end loop

The final sanity check: confirm the dashboard sees a state change in real time.

1. Pick an endpoint that currently reports HEALTHY in the dashboard
2. Stop the BES Client on that endpoint
3. Within 30 to 60 seconds the watchdog daemon should detect the stop and fire a force-now run
4. The dashboard should flip the endpoint to UNHEALTHY within roughly 30 seconds of the BES Client stopping
5. Start the BES Client again. The dashboard should flip back to HEALTHY on the next heartbeat

Troubleshooting

Common installation and post-install issues, with the fastest path to diagnosis.

Dashboard Server

Symptom	Cause	Resolution
Service fails to start	Varies; the journal names the exact error	sudo journalctl -u bigfix-health-archive -n 100
502 Bad Gateway in the browser	App not running, OR warmup window not yet complete	Wait 60 seconds; if still failing, check journalctl
BES Cross-Check returns 0 endpoints	Operator is not a Master Operator	Re-create the BigFix operator with MO role
BES Cross-Check fails with SSL error	Tenant Root CA not uploaded, or expired	Re-capture or re-paste the Root CA: Administration > System Settings > Connections, select the tenant, BES ROOT CA CERTIFICATE card
SSL certificate warning in browser	Self-signed cert (expected)	Replace with a trusted cert in Step 7 of Part 1
Cannot reach BigFix server from dashboard	Network or firewall block	curl -sk https://bigfix-server:52311 from the dashboard host
Database connection error	PostgreSQL not running	sudo systemctl status postgresql
Hostname change broke credentials	.env credentials are encrypted under a key derived from machine-id + hostname; after a rename the ciphertext no longer decrypts	cd /opt/bigfix-health-archive && sudo venv/bin/python -m app.scripts.rotate_credentials --old-hostname OLD, then restart both services. Works even after a failed restart; a timestamped .env backup is taken automatically.

Health Agent (general)

Symptom	Cause	Resolution
Fixlet runs, endpoint not visible in dashboard	Charter invalid or missing	Re-stage a valid dashboard.charter, or supply the URL and key when prompted
Endpoint shows REVOKED	Identity was revoked from the dashboard side	Select the endpoint and click Restore (in the Endpoints list bulk bar or the endpoint detail panel); it re-enrolls cleanly on its next post
Endpoint shows TAMPERED integrity state	Binary SHA-256 does not match the build attestation	Confirm the bundle on the Console workstation matches the distribution checksum; redeploy
Endpoint shows CHAIN INVALID	Code-signing cert not in the endpoint's Trusted Publisher store	Run the Trust-Bootstrap Fixlet against this endpoint
Endpoint reports SILENT despite being online	Agent not firing on its timer	Check the OS scheduler entry; see the platform-specific tables below

Health Agent (Linux specific)

Symptom	Cause	Resolution
Timer not firing	Timer was not enabled	systemctl enable --now agentmonitor.timer
Agent crashes on first run	identity.json owned by wrong user	chown agentmonitor:agentmonitor /var/lib/bigfix-health-agent/identity.json
CRYPTO_BROKEN.txt present in /var/lib/bigfix-health-agent/	Runtime crypto self-check failed at startup	Reinstall the agent with the install directory excluded from AV real-time scanning; the install-time crypto check reports which submodule is missing
Agent service fails (exit status 2), log shows enrollment deferred and 401	BES Client not installed or not yet reporting on this endpoint	Install the BES Client; the agent enrolls and reports on its next timer run

Health Agent (Windows specific)

Symptom	Cause	Resolution
Install fails with ModuleNotFoundError for cryptography	AV interrupted the bundle extraction (partial extract regression)	Stop AV scanning of the install dir, re-take action
AgentMonitor.exe blocked by SmartScreen	Code-signing cert not yet trusted on this endpoint	Run the Trust-Bootstrap Fixlet first
Scheduled task missing or disabled	Install partially failed	Check the Fixlet output; re-take action
regapp inspector does not see the application	By design: the installer registers scheduled tasks, not an Add/Remove Programs (HKLM Uninstall) entry	Use a BigFix version-of-file inspector against AgentMonitor.exe, or Task Scheduler, to confirm the agent is installed
AgentMonitor not visible in Add or Remove Programs	Same as above	Use Task Scheduler to confirm the agent is installed

Log file locations

Component	Log location
Dashboard application	sudo journalctl -u bigfix-health-archive
Dashboard Nginx access	/var/log/nginx/bigfix-health-archive-access.log
Dashboard Nginx error	/var/log/nginx/bigfix-health-archive-error.log
Dashboard installer	install.log (in the extracted tarball directory)
Linux Health Agent	/var/lib/bigfix-health-agent/agent.log
Linux Watchdog daemon	sudo journalctl -u agentmonitor-watchdog.service
Windows Health Agent	C:\Program Files\AgentMonitor\agent.log
Windows installer (Fixlet)	BigFix action log on the endpoint

Managing the Application

Service management (Dashboard)

bash

```bash
# Check status
sudo systemctl status bigfix-health-archive

# Restart the service
sudo systemctl restart bigfix-health-archive

# Stop the service
sudo systemctl stop bigfix-health-archive

# View recent logs
sudo journalctl -u bigfix-health-archive -n 200

# Follow logs in real time
sudo journalctl -u bigfix-health-archive -f
```

The dashboard also runs a separate scheduler service, bigfix-health-archive-scheduler, which runs the silent detector, alerting, and the BigFix cross-check. Restart it alongside the main service after an upgrade.

Database backup (from the web UI)

1. Navigate to **Administration > System Settings > Backup and Restore**
2. Click **Create Backup**
3. Download the resulting .sql file for off-site storage

Backup format. Backups are plain pg_dump .sql files stored under /opt/bigfix-health-archive/backups/. Restore is destructive (it drops and recreates the database); a pre-restore safety backup is always taken automatically first, so you can roll back.

Upgrading the dashboard

When a new dashboard release ships, you will receive a new tarball. The installer detects an existing install and switches to upgrade mode:

1. Extract the new tarball into a fresh directory
2. Run `sudo bash install.sh` as before
3. The installer preserves your .env (BigFix credentials, SMTP creds, app secret), the SSL certificate, the admin password, and all of the database contents
4. Application code is replaced; database migrations are applied automatically

Upgrading the Health Agent

Agent upgrades use the same Fixlet you used for first-time deployment. When a new agent version ships:

1. Replace the bundles on your Console workstation with the new versions
2. Update the Fixlet's version number in the .bes XML (or import the new .bes shipped with the release)
3. Take action against your target computer group
4. The Fixlet's relevance gate only fires against endpoints below the new version, so re-taking action is safe

Linux Health Agent management (per endpoint)

bash

```bash
# Service status
systemctl status agentmonitor.timer
systemctl status agentmonitor-watchdog.service

# Force an immediate agent run (bypasses the self-check interval)
```

```
sudo /opt/agentmonitor/agentmonitor \
    --config /etc/agentmonitor/config.ini --force-now

# View the agent log
sudo tail -f /var/lib/bigfix-health-agent/agent.log
```

Windows Health Agent management (per endpoint)

PowerShell

```
# Task status
Get-ScheduledTask -TaskName "AgentMonitor"
Get-ScheduledTask -TaskName "AgentMonitor-Watchdog"

# Force an immediate heartbeat
Start-ScheduledTask -TaskName "AgentMonitor"

# Disable temporarily
Disable-ScheduledTask -TaskName "AgentMonitor"
Enable-ScheduledTask  -TaskName "AgentMonitor"

# View the agent log
Get-Content "C:\Program Files\AgentMonitor\agent.log" -Tail 50 -Wait
```

Uninstalling

To remove the dashboard, run sudo bash uninstall.sh from the extracted bundle. It stops and removes the services, removes the Nginx site, drops the database and role, and removes the application directory, TLS certificate, internal CA, sudoers rule, and service account. Add **--keep-db** to preserve the database and its data for a future reinstall, or **--dry-run** to preview the changes without removing anything. PostgreSQL and Nginx themselves are left installed.

To remove the Health Agent from an endpoint: on Linux, run sudo bash install.sh **--uninstall** from the extracted bundle; on Windows, run AgentMonitor-Installer.exe **--uninstall** as Administrator. Either can be deployed at scale through a BigFix Fixlet. The agent posts a goodbye on the way out and the endpoint hides itself from the dashboard automatically.